CARD 1 OF 8 THE 18-INCH RULE **QUESTION**

Standard **pendent or upright** sprinklers over stock. What's the minimum clearance, and where is it measured **from**?

FSM · Fire System Mastery firesystemmastery.com

CARD 2 OF 8 TWO CLEARANCES **QUESTION**

A storage design table lists **"clearance to ceiling."** Is that the same as the 18 in. clearance to the deflector?

FSM · Fire System Mastery firesystemmastery.com

CARD 3 OF 8 ESFR **QUESTION**

ESFR sprinklers over storage. What's the minimum clearance from deflector to top of storage?

FSM · Fire System Mastery firesystemmastery.com

CARD 4 OF 8 CMSA **QUESTION**

Which other sprinkler type carries a **36 in.** storage clearance, and what's the catch with the pendent version?

FSM · Fire System Mastery firesystemmastery.com

CARD 2 OF 8 TWO CLEARANCES **ANSWER**

No. They're two different measurements.

Clearance to ceiling: top of storage up to the **ceiling**. Storage designs use it to pick protection.

Clearance to storage: top of storage up to the **deflector**. That's the 18 in. (or 36 in.) that keeps the spray pattern open.

NFPA 13 (2019) · Ch. 3 · Ch. 10

On the job: a 30 ft ceiling over 25 ft of stock is 5 ft to the ceiling. The deflector gap is smaller than that.

FSM · Fire System Mastery firesystemmastery.com

CARD 1 OF 8 THE 18-INCH RULE **ANSWER**

18 in. or more, measured from the sprinkler **deflector** down to the top of the storage.

If another NFPA standard asks for **more** clearance for that hazard, the bigger number wins.

Annex: the 18 in. isn't meant for vehicles in concrete parking garages.

NFPA 13 (2019) · Ch. 10 · Annex A

Trap: measuring 18 in. from the **ceiling**. With a 10 in. drop, that leaves only 8 in. under the head.

FSM · Fire System Mastery firesystemmastery.com

CARD 4 OF 8 CMSA **ANSWER**

CMSA (control mode specific application) sprinklers: **36 in. or more** from the deflector to the top of storage.

The catch: the CMSA section is written around uprights. **CMSA pendents** follow the general sprinkler rules and the manufacturer's installation instructions instead.

NFPA 13 (2019) · Ch. 13

On the job: on a CMSA pendent job, pull the data sheet for the clearance. Don't assume.

FSM · Fire System Mastery firesystemmastery.com

CARD 3 OF 8 ESFR **ANSWER**

36 in. or more. Twice the standard-spray number.

ESFR is built to **suppress** a fire, not just control it. It needs a big, fast, fully formed discharge reaching the top of the pile. Crowd it and you lose the very thing the design counts on.

NFPA 13 (2019) · Ch. 14

Trap: "We're over 18 in., we're fine." Under ESFR, 30 in. of clearance is a deficiency.

FSM · Fire System Mastery firesystemmastery.com

sheet 1 BACK — columns are intentionally mirrored for long-edge duplex · card numbers must match front to back

CARD 5 OF 8 SIDEWALL & EC **QUESTION**

Does the storage clearance change for **sidewall** or **extended coverage** sprinklers?

FSM · Fire System Mastery firesystemmastery.com

CARD 6 OF 8 WALL SHELVING **QUESTION**

A stockroom has shelving along the walls. Can the top shelf go **higher** than 18 in. below the deflectors?

FSM · Fire System Mastery firesystemmastery.com

CARD 7 OF 8 STORAGE CHAPTER **QUESTION**

In the **storage** chapters, when can the clearance be **less** than 18 in.? And which commodity **always** needs 36 in.?

FSM · Fire System Mastery firesystemmastery.com

CARD 8 OF 8 WHY 18 INCHES **QUESTION**

Why **18 in.**? What happens in that space under a sprinkler?

FSM · Fire System Mastery firesystemmastery.com

CARD 6 OF 8 WALL SHELVING **ANSWER**

Depends where the shelf sits.

On the wall, not directly under a head: yes. The 18 in. plane doesn't cap wall shelving.

Directly under a sprinkler: no. The shelf and everything on it has to stay below the 18 in. plane.

NFPA 13 (2019) · Ch. 10

On the job: a head lands right over a tall wall unit. Move the head or drop the top shelf.

CARD 5 OF 8 SIDEWALL & EC **ANSWER**

No. Still 18 in. or more from the deflector to the top of storage for:

- Standard sidewall
- Extended coverage upright and pendent
- Extended coverage sidewall

The jump to **36 in.** is for ESFR, CMSA, and the special cases in the storage chapter.

NFPA 13 (2019) · Ch. 10 · Ch. 11

Trap: the EC pendent in a stockroom gets the **same** 18 in. as a standard head, not more.

CARD 8 OF 8 WHY 18 INCHES **ANSWER**

That's where the spray pattern forms. NFPA 13 treats the first 18 in. below a deflector as the zone where an obstruction can keep the pattern from fully developing.

Storage piled into that zone acts like an obstruction. The water hits the top box instead of spreading over the coverage area.

NFPA 13 (2019) · Ch. 9 · Ch. 10

On the job: the clearance was fine at turnover. Six months later the stock is stacked higher. Walk the floor and check it.

CARD 7 OF 8 STORAGE CHAPTER **ANSWER**

Less than 18 in.: only where **large-scale fire tests** have proven it for that particular hazard.

Always 36 in.: rubber tires.

Special sprinklers may also be set up with a 36 in. clearance. And if another standard demands more, follow it.

NFPA 13 (2019) · Ch. 20

Trap: treating tire storage like any other stock at 18 in. It's 36 in. no matter what head is used.